AzureML Cloud Cost Evaluation Report

1. Introduction

This document estimates the financial costs associated with deploying and running the AxonRooter pipeline on Azure Machine Learning (AzureML) and Portainer according to the Pricing Calculator provided by Microsoft. The evaluation is calculated based on data storage, model training and inference, retraining with user feedback data, Kubernetes cluster usage, monitoring and logging. This document also analyses considerations such as scalability and cost-saving strategies.

2. Storage Costs

The project utilized an initial dataset of approximately 2 GB stored in Azure Blob Storage. The rate of our standard HDD disk from 4 to 32GB is €1.54 per month (standard tier in West Europe). Due to our user feedback feature which allows the users upload new image and mask patches to our database for model retraining, we expect increasing data volume over time, hence we will upgrade to 64GB, 128GB or larger plans on demands, but a reasonable estimation will involve 128GB at most.

Thus, the estimated cost of this section ranges from €1.54 (4-32GB) to €5.89 (10-128GB) per month.

3. Kubernetes Cluster Usage

The AzureML compute cluster was configured using Kubernetes. The costs are based on the CPU, memory, and GPU allocations as below:

Model Training Costs

Model training will be executed on AzureML on a standard NC6 GPU virtual machine (VM) consisting of 1 x NVIDIA Tesla K80 GPU, 6 vCPU and 56 GB RAM, as our dataset is relatively small while increasing, higher plans such as A100 would be overkill for our project. Each training run lasted approximately 0.8 hours and ran approximately 10 times for fine-tuning the model and cross validation.

- - Rate: €1.032/hour (West EU rate)
  - Runtime: 0.8 hours
  - Total training cost: 0.8 * 10 * 1.032 = €8.216

- Retraining costs: Additionally, the pipeline is designed to accept updated image submissions from users. These new samples are integrated into the dataset, and a retraining cycle is scheduled on a weekly basis. This ensures the model remains up-todate with the latest data and improves generalization to new cases. Weekly retraining will increase cumulative compute costs depending on the volume and duration of each session. The training time will be gradually increasing due to continuous data supplement. The retraining will be executed on the same VM as the initial training:

- Rate: €1.032/hour (West EU rate)
  Runtime: 0.8 – 2 hours/per retraining process
  Retrain times per month: 4 * 10
  Final cost for retraining: €33.02 – 82.56/month

Inference Costs

Inference operations (evaluation and registration steps) were performed on a Standard_D2s_v3. Each inference run is assumed to take about 0.04 hours, and used by 20 users/day, each triggering 5 inferences:

- - Rate: €0.096/hour (West EU rate)
  - Runtime: 0.04 hours * 20 * 5 * 30 = 120 hours/month
  - Total inference cost: €11.52/month

Load Balancer

The Kubernetes load balancer provides a public IP for our application, and automatically distributes external traffic to the pods evenly across multiple nodes.

Rate: €0.022/hour

Total cost: €0.022 × 24 × 30 ≈ €15.84/month

4. Monitoring and Logging

Application Insights and Azure Monitor are used for real-time monitoring and logging. Although typically inexpensive, costs can grow with high-frequency logging and long retention periods.

- Estimated monitoring cost: €5–10/month depending on usage volume

5. Frontend Deployment

The frontend of the project is built using Streamlit, a lightweight Python framework for building interactive web applications. It serves as the user interface for interacting with the backend machine learning model deployed via Azure ML. The app is hosted on Portainer on local server provided by Buas as far as we planned, which is for free. If our application were to acquire sufficient investment or Buas stops providing this service, we will host on a lightweight VPS with 2 vCPU / 4 GB RAM, which can handle more users and heavier data processing. It will cost €10 – 12 per month.

6. Total cost

The total cost for our application from backend to frontend deploying and maintenance is all listed above, summed up to a range from:

Minimum: €76.92 per month + €8.216 (fixed initial training cost)

To:

Maximum: €137.81 per month + €8.216 (fixed initial training cost)

7. Cost saving strategies

1. To balance performance, reliability, and budget constraints, we set reasonable VM requirements tailored to our specific scenarios. Our dataset is considered small to medium size, with low user requests as an unpublished student project, therefore we chose an HDD disk with minimal storage to store our dataset and AzureML over large storage SSD ones. Our CNN model isn't too deep, which doesn't require extreme compute resources. Hence, we use a single Nvidia Tesla K80 with 6 vCPU and 56 GB RAM to be capable of our tasks and doesn't waste unnecessary energy and costs.

2. The inference and frontend procedure requires minimal CPU and memory, for the concerns of budget, performance and feasibility for our group without HTML/CSS experience, we compiled our frontend on Streamlit. The application was containerized and hosted via Portainer, a lightweight and currently cost-free solution for our team, ensuring ease of deployment while minimizing operational overhead and infrastructure costs.

3. Monitoring and logging are not maturely deployed in our backends due to limited time, but we have set complete logging sections inside our Streamlit application which display, and store detailed technical information for users and developers. We plan to use Portainer's built-in monitoring UI to view resource usage of individual containers, which is also free but fully capable of our tasks.